1. Basics of dg modules.

① Two definitions:

- M is a graded A-mod. s.t. $\partial_M(am) = \partial_A(a)m + (-1)^{|a|} a\,\partial_M(m)$.
- M is a complex with a map of complexes: $\Updownarrow$

$\begin{cases} \mu_M : A \otimes_k M \to M,\ a \otimes m \mapsto am & \text{since this is exactly } M(\partial_{A\otimes M}(a\otimes m)) \\ \text{that is unital}: & \text{conditions for } A\text{-mod.} \\ \& \text{ associative}: & \text{conditions for } A\text{-mod.} \end{cases}$

② Facts

i) If M dg A-mod., then $H(M)$ a graded $H(A)$-mod.

ii) Def: $A^\natural$ & $M^\natural$ the underlying graded Q-mod. (without differentials).

$A^\natural$: graded Q-alg. $M^\natural$: graded $A^\natural$-mod.

iii) E-mod.: $E := Q[e_1, \dots, e_n : \partial(e_i) = f_i]$.

Then a dg E-mod. F amounts to R-lin. maps $\sigma_1, \dots, \sigma_n \in \mathrm{Hom}_R(F,F)$ s.t.:

$$\underbrace{\sigma_i\sigma_j = -\sigma_j\sigma_i,\ \sigma_i^2 = 0}_{\text{exterior conditions}},\quad \underbrace{\partial\sigma_i + \sigma_i\partial = f_i\,\mathrm{id}_F}_{\partial(e_i\cdot x) = f_i x - e_i\cdot\partial(x)}$$

and $e_i \cdot x = \sigma_i(x)$.

✱ e.g. $Q = k[[x,y]]$, $f = x^3 + y^4$, $E = Q[e \mid \partial(e) = f]$, F is:

$$0 \to Q \xrightarrow{\begin{bmatrix} -y \\ x \end{bmatrix}} Q^2 \xrightarrow{[x,y]} Q \to 0.$$

To give a Q-mod. structure $\Leftrightarrow$ to give nullhomotopy for $f_\cdot$.

Can check:

$$0 \to Q^2 \xrightarrow{\begin{bmatrix} -y \\ x \end{bmatrix}} Q^2 \xrightarrow{[x,y]} Q \to 0$$

$$\sigma_2 = [x^2 \ -y^2] \qquad \begin{bmatrix} x^2 \\ y^2 \end{bmatrix} = \sigma_1$$

(Note: σ_i's have one component of σ)

is a nullhomotopy for f.

③ Tensor & Hom.

Note that dg A-mod has two structures:

graded A-mod structure & differential structure

A-linear maps & dg-mod. maps.

and on dg A-mod. M, they're compatible.

For tensor & Hom, only take care of the graded A-mod. structure, and:

- for tensor, the differential is inherited;
- for Hom, Z_0 is exactly the dg A-mod. maps.

i) Def: $M \otimes_A N := M \otimes_Q N / (am \otimes n - (-1)^{|a||m|} m \otimes an)$

$\partial_{M \otimes_A N} = \partial_{M \otimes_Q N}$ here, and it's well defined on ↘ (can check it's 0).

e.g. Consider $A := Q[x]$, M an A-mod. Then $R = Q[x]/I+(x)$.

Thus: $M \otimes_{Q[x]} R = M/(I+(x))M$.

ii) Def: $Hom_A(M,N) := \{ a \in Hom_Q(M,N) : a(am) = (-1)^{|a||a|} a\, a(m) \}$.

$\partial_{Hom_A(M,N)}(a) = \partial^N \circ a - (-1)^{|a|} a \circ \partial^N$. (same as in Hom_Q).

Rmk: Def of map of dg A-mod., homotopy & quasiisomorphism are exactly the same as in graded Q-mod., since $\partial_{Hom_A(M,N)}(a)$ is the same.

Then Z_0, H_0 of $Hom_A(M,N)$ is the same.

iii) Def: $(\Sigma^l(M))_n := M_{n-l}$, $\partial^{\Sigma^l M} := (-1)^l \partial^M$, with A-mod structure:

$$a \cdot \Sigma^l m := (-1)^{|a| \cdot l} \Sigma^l (am)$$

↓ basically differ from "true" am by a sign.

Note all of these are generalisations R-mod., since we let $A_0 = R$, $A_i := 0 \ \forall i \neq 0$.

④ Cone.

i) Def: given $\varphi : M \to N$ of dg A-mod., the mapping cone is:

$N \oplus \Sigma M$ with $\begin{bmatrix} \partial^N & \varphi \\ 0 & -\partial^M \end{bmatrix}$.

ii) Properties

- $0 \to N \xrightarrow{\alpha} cone(\varphi) \xrightarrow{\beta} \Sigma M \to 0$.

  α & β are indeed chain maps: the diagonals are ∂^N & $\partial^{\Sigma M}$.

- φ quasiiso $\Leftrightarrow$ cone(φ) exact.

Pf: use LES of the above.

• φ is nullhomotopic $\Leftrightarrow$ φ factors through the canonical $M \to \mathrm{cone}(\mathrm{id}_M)$.

Pf: Consider $M \xrightarrow{i} M \oplus \Sigma M \xrightarrow{\frac{\varphi+\sigma}{g}} N$, with φ the composite.

"$\Rightarrow$": Then we define $\mathrm{cone}(\mathrm{id}_M) \to N$ as $\varphi + \sigma$

"$\Leftarrow$": Then we define $\sigma(m) = g(sm)$.

& the differentials agree by nullhomotopy & dg mod. map.

2. Semifree & semiprojective.

We only discuss bounded below dg modules in this section.

Def: F is <u>semifree</u> if it admits a filtration by dg A-submodules

$$0 = F(-1) \subset F(0) \subset F(1) \subset \cdots \subset F$$

s.t. $F(i)/F(i-1) = \oplus \Sigma^{|d|} A$.

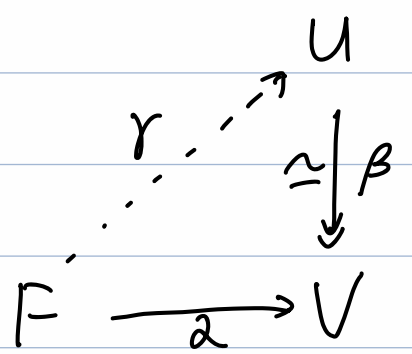
✡✡ Note (dg, bded.) F is semifree $\Leftrightarrow$ $F^\natural$ is free graded $A^\natural$-module,

i.e. $F = \bigoplus_{\lambda \in \Lambda} \Sigma^{|e_\lambda|} A$.

Here $F(d) = \bigoplus_{|e_\lambda| \le d} \Sigma^{|e_\lambda|} A$, and we denote the basis as $\{e_\lambda\}_{\lambda \in \Lambda}$.

↓ assume bded below at 0.

Def: P is <u>semiprojective</u> if



γ is unique up to homotopy.

① Thm: Every M admits a semifree res. over A, unique up to hom. equiv.
(semifree F s.t. $F \simeq M$)

pf:

i)

- We need: Z^0 a generating set of $Z(M)$,

$$F^0 := \bigoplus_{z \in Z^0} \Sigma^{|z|} A \quad \text{attaching } X^0 \text{ (new variables).}$$

$$\varepsilon^0 : F^0 \to M$$

$$x_z^0 \mapsto z.$$

- Then $H(\varepsilon^0) : H(F^0) \to H(M)$ is surjective.

ii) Induction step: we need to kill the cycles in $\ker(\varepsilon^i)$.

- For $H(\varepsilon^i) : F^i \to M$, fix Z^i whose homology class is the generating set for $\ker(H(\varepsilon^i))$

✡ $\forall z \in Z^i$, $H(\varepsilon^i)(z) = [\varepsilon^i(z)] = 0$ in $H(M)$

$\Rightarrow \varepsilon^i(z) = \partial^M(m_z)$

$$z \in Z^i \quad \varepsilon^i(z) = \partial^M(m_z) \quad \longleftrightarrow \quad m_z \in M$$

- ✡ $F^{i+1} := \text{cone}\left(\bigoplus_{z \in Z^i} \Sigma^{|z|} A \xrightarrow{\bigoplus (x_z^i \mapsto z)} F(i) \right)$

this is "attaching" the new variables !!

they're the new variables

and

✡ on the new variables, $\varepsilon^{i+1}(x_z^i) = m_z$ (so they kill the cycles since x_z^i becomes the boundary $\partial x_z^i = z$)

- $\begin{cases} \varepsilon^{i+1} \text{ extends } \varepsilon^i \text{ since } \partial^M(m_z) = \varepsilon^i(z) \\ F(i+1)/F(i) = \bigoplus_{z \in Z^i} \Sigma^{|z|+1} A \quad \text{since the cone shifts } \bigoplus \Sigma^{|z|} A. \end{cases}$
- $F := \varinjlim F(i)$, $\varepsilon = \varinjlim \varepsilon(i)$, now:

a) F is semifree by construction

b) $H(\varepsilon)$ is surjective since it extends ε^0; $\ker(H(\varepsilon)) = 0 \Rightarrow \varepsilon$ quasiiso.

c) ε is surjective since ε is surjective on cycles & homology.

Rmk:

i) Semifree is usually adjoining new variables on cycles.

e.g. to construct F with $F(i)/F(i-1) \cong \Sigma^i A$, we adjoin one variable e_z s.t. $\partial(e_z) = z$ each time.

ii) If $H_0(M)$ Noeth. & $H(M)$ f.g. over $H_0(M)$, then res. is f.g. at each stage:

$$F^q \simeq \bigoplus_{i=n}^{\infty} \Sigma^i (A^q)^{\oplus b_i}.$$

Equivalently,

$$F^q \simeq A^q \otimes_{\mathbb{Q}} G$$

↓

G bded below graded $\mathbb{Q}$-mod., G_i finite rank free.

② Semifree $\Rightarrow$ semiprojective.

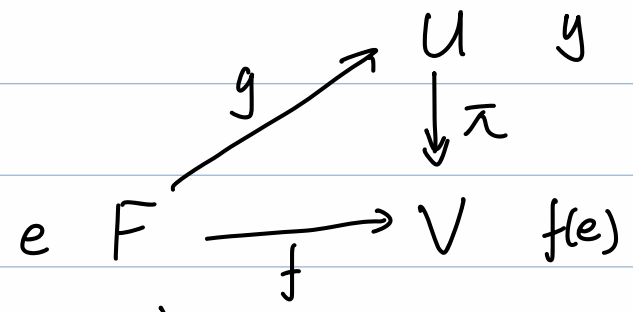
pf: See Avr, p. 13.

We only sketch the existence part.

- We use induction on $F(n)$.

Assume for $F(n)$ we have:

s.t. $\pi g = f$.



Now for any new variable $e \in F(n+1)$,

π surjective $\Rightarrow \exists\, y \in U$ s.t. $f(e) = \pi(y)$.

- Measure the failure of differential:

$c := \partial y - g\partial(e)$; $\partial(e) \in F(n)$, so $\pi(c) = \pi\partial y - \pi g\partial(e) = \partial f - f\partial = 0$.

"$g(e)$" ; $\partial y = \partial f(e)$; f on $F(n)$

- Thus $c \in \ker\pi \xrightarrow{\pi \text{ quasi-iso}} c = \partial h$. Now let $g(e) = y - h$.

So g extends on $F(n)$: $\begin{cases} \pi g = f \\ \partial g = g\partial \end{cases}$

Thus g extends on all F.

③ Properties of semiprojectivity ☆.

Thm: P dg A-mod. TFAE:

i) P is semiprojective

ii) For any surjective quasiisom. $M \to N$,

$$\beta_* = \mathrm{Hom}_A(P, \beta) : \mathrm{Hom}_A(P, M) \to \mathrm{Hom}_A(P, N)$$

is also surjective quasiisom.

iii) $P^\flat$ proj. and $\mathrm{Hom}_A(P, C) \simeq 0$ if $C \simeq 0$

iv) dg mod. P is direct summand of a semifree dg A-mod.

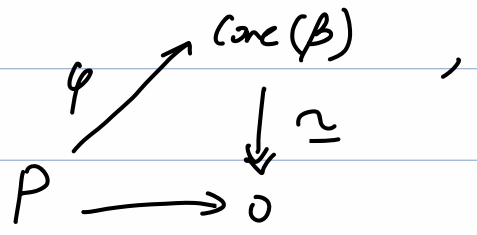
pf:

i) ⇒ ii):

- Preserves surjectivity: by definition.
- Preserves quasiisom.: first, $\mathrm{cone}(\mathrm{Hom}_A(P,B)) = \mathrm{Hom}_A(P, \mathrm{cone}(B))$

since $\mathrm{Hom}_A(P,-)$ preserves direct sums.

Then for any $P \xrightarrow{\varphi} \mathrm{cone}(B)$, $\exists$



,

So φ is homotopic to 0 which also lifts $\simeq$, so $\mathrm{cone}(B_*) = 0$

ii) ⇒ iii): clear by preserving surjectivity & quasiisom.

iii) ⇒ iv):

- P has a semifree resolution $\varepsilon: F \xrightarrow{\sim} P$.

$P^\natural$ projective $\Rightarrow \exists\, s: P^\natural \to F^\natural$ s.t. $\varepsilon s = id_P$ since the SES ending with $P^\natural$ splits.

- Since ε quasiiso., $K := \ker(\varepsilon) \simeq 0 \Rightarrow \mathrm{Hom}_A(P,K) \simeq 0$.
- s may not be chain map, let $\partial(s) = \partial_F \circ s - s \circ \partial_P$, which is precisely ∂ on $\mathrm{Hom}_A(P,K)$: note $\varepsilon\partial(s) = \partial_P - \partial_P = 0$.
- $\partial^2 = 0$, so $\partial(s)$ is a cycle, and $\mathrm{Hom}_A(P,K) \simeq 0 \Rightarrow \partial(s) = \partial(h)$.

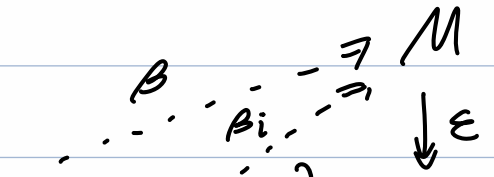
Now let $s' := s - h$, where $h \in \mathrm{Hom}_A(P,K) \Rightarrow \varepsilon s' = id_P$.

↓ this says $\varepsilon h = 0$

iv) ⇒ i): From ② we know F is semiprojective, where

$$P \xrightarrow{i} F \xrightarrow{\pi} P.$$

Given a diagram



we have:

Now $\varepsilon\beta i = \alpha\pi i = \alpha$, βi unique up to homt. since $\pi\beta i$ is.

④ Minimal semifree resolution.

Def: $F \xrightarrow{\sim} M$ is minimal if $\partial(F) \subset m_A F$, where m_A max. of A.

Note this is similar to minimal free res., rather than minimal model.

Prop: $\exists$ a minimal semifree res. for $H(M)$ f.g. over $H(A)$; (A,m) local.

Pf: We show that we can always make F minimal.

- Assume F is not minimal. Then $\exists$ homogeneous basis elt. $e \in F$ s.t.

$$\partial(e) = \sum a_f f$$

where for some a_f, $a_f \notin m$.

- Then a_f is unit, and $f' := \sum a_f f$ we have $f = a_f^{-1}(f' - \sum a_g g)$, and this is a linear change of basis $\Rightarrow f'$ is the new basis elt.

- Now $Ae \oplus Af \cong \mathrm{cone}(id_A)$ since $\underbrace{\partial(e) = f,\ \partial(f) = 0}$.
  what determines $\mathrm{cone}(id_A)$.

$\Rightarrow F = G \oplus (Ae \oplus Af)$ and $F \simeq G$

(• Alternatively, B.7 in Avr-Hote).

(Or we can use NAK & ∂ is an isomophism on v.s.).

⑤ Properties of semifree F.

i) Thm: $\mathrm{Hom}_A(F,-)$ and $F\otimes -$ preserves quasiisom.

pf: See Prop 1.3.2 in Avr.

ii) Change of basis.

Thm: $A \simeq B$ quasiisom. of dgas. M a dg B-mod.

- $\varepsilon : F \to M$ is a semifree resolution of dg A-modules $\Leftrightarrow$ $B\otimes_A F \xrightarrow{id_B\otimes\varepsilon} B\otimes_A M \xrightarrow{\mu} M$ is a semifree res. over B.
- Additionally, if $H(M)$ f.g. over $H(B)$, then F minimal semifree A-res. $\Leftrightarrow$ $B\otimes_A F$ min. semifree B-res.

3. ✡ Important result.

Thm: Let $\widehat{R} \cong Q/I$ be min. reg. repn., $A := Q[X]$ a min. model for R. Then $A \xrightarrow{\sim} \widehat{R}$.

For f.g. R-mod. M, if $F \xrightarrow{\sim} \widehat{M}$ is a min. semifree A-res., with $F^q \cong \bigoplus_{i\geq 0} \Sigma^i (A^{\beta_i})^q$, then:

$$\mathrm{pdim}_R(M) < \infty \Leftrightarrow \beta_i = 0 \text{ for } i \geq n \text{ for some } n,$$

and $\beta_i = \beta_i^R(M) = \mathrm{Tor}_i^R(M,k)$

↓

Betti no.

In short: min. semifree $Q[X]$-mod. $\simeq \hat{M} \Leftrightarrow$ min. free $\simeq M$.

Pf: We use 2.⑤. ii) with $A = Q[X]$, $B = \hat{R}$.

- $F \xrightarrow{\sim} \hat{M}$ is a min. semifree $Q[X]$-res., then by 2.⑤. ii):
  $B \otimes_A F \xrightarrow{\sim} \hat{M}$ is a min. semifree B-res. of $\hat{M}$.
- ✡ B is concentrated on deg. 0, so
  $$B \otimes_A F = \bigoplus_{i \geq 0} \Sigma^i (\hat{R})^{\beta_i}$$ is a min. free $\hat{R}$-res. of $\hat{M}$.
- ✡ $\mathrm{pdim}_{\hat{R}}(\hat{M}) = \mathrm{pdim}_R(M)$ since $\hat{R}$ is ff. (consider $F \otimes_R \hat{R}$).
  Thus $\mathrm{pdim}_R(M) < \infty \Leftrightarrow$ minimal semifree $Q[X]$-res. of $\hat{M}$ is finite.

✡✡ Rmk: the point of this theorem is study $Q[X]$-res., which contains more singularity info. (it records singularity)

e.g. $Q = k[x,y]$, $R = Q/(xy)$, and a min. free res. for k is:

$$\cdots \to R^2 \xrightarrow{[x\ y]} R^2 \xrightarrow{[y\ x]} R^2 \xrightarrow{[x\ y]} R \to k$$

& $Q[X] = Q \oplus Qe$, $\partial(e) = xy$.

To obtain $Q[X]$-res, we can lift this to the regular Q and "correct" the singularities.

$$\begin{array}{ccccccccc}
\cdots \to & Q^2 & \xrightarrow{[x\ y]} & Q^2 & \xrightarrow{[y\ x]} & Q^2 & \xrightarrow{[x\ y]} & Q & \to k \\
 & \oplus & \searrow^{-e I_2} & \oplus & \searrow^{[-1\ -1]} & \oplus & \nearrow_{xy} & & \\
\cdots \to & Q^2 & \xrightarrow{[y\ x]} & Q^2 & \xrightarrow{[x\ y]} & Qe & & &
\end{array}$$

(xy, xy) must be corrected by $-(\partial(e), \partial(e))$.

Here $(a+b)xy$ must be corrected by $-(a+b)\partial(e)$.

This F is indeed min. semifree.

& this verifies the Thm.